# Predicting Cross-Source Matches of Noisy Business Records: Logistic, Discriminant and Count Models for Entity Resolution

Manthan Vats
Reg. No. 2427030314
Department of Computer Science and Engineering
Manipal University Jaipur, Jaipur, India
manthan.ralph17@gmail.com

Peddi Reddy Srikar Reddy
Reg. No. 2427030741
Department of Computer Science and Engineering
Manipal University Jaipur, Jaipur, India

***Abstract*—Records of the same business often sit in several databases under different spellings, abbreviations and scripts, with no shared identifier. We treat this entity resolution task as a predictive modeling problem on the Amazon ML Challenge 2026 data, which links US and Indian business records across three sources. TF-IDF blocking reduces each query to about 50 candidates, and every candidate pair is described by 43 engineered features. On a 5% stratified sample we compare logistic regression, linear and quadratic discriminant analysis and gradient boosting (LightGBM), tuned by grid and random search under five-fold grouped cross-validation and tested on a lockbox of 21,685 entities. LightGBM reaches a macro F0.5 of 0.9855 and logistic regression 0.9716, while LDA and QDA trail at 0.948 and 0.958. The logistic coefficients show that address and house-number agreement, together with competition between entities for the same record, predict a match better than name similarity. An expected-F0.5 decision rule beats fixed probability thresholds for calibrated models. We also model the number of matches per entity: the counts are under-dispersed (Pearson dispersion 0.31), so the negative binomial model collapses to the Poisson model, and a multinomial logit classifies the counts more accurately. The full version of the pipeline scored 0.959 on the challenge public leaderboard.**

***Index Terms*—entity resolution, record linkage, logistic regression, discriminant analysis, gradient boosting, count regression, F-measure**

## I. Introduction

Marketplaces, banks and logistics firms keep records of the same businesses in several systems, and those systems rarely share an identifier. One source lists "Sharma Traders Pvt Ltd, Plot 14, MG Road, Pune", a second writes "Sharma Traders Private Limited, 14 Mahatma Gandhi Rd" and a third gives the name in Devanagari with no PIN code. Deciding which records belong to the same real business is called entity resolution (ER) or record linkage [1], [2]. Mistakes are expensive in both directions: a missed link splits one seller into two accounts, while a false link merges the payment history of two unrelated companies.

We study ER as a predictive modeling problem. For each record of a reference source (Source 1) we must predict the set of records in two other sources that describe the same business. The data is the training set of the Amazon ML Challenge 2026 business entity resolution task: 2.2 million Source 1 records from the US and India, 10.3 million Source 2/3 records and 7.6 million labelled matches. Our team took part in the challenge with a two-stage gradient boosting system that reached a public leaderboard score of 0.959 (macro F0.5). For this paper we return to the problem with the tools of a predictive analytics course and ask three questions. How far do interpretable classifiers (logistic regression, linear and quadratic discriminant analysis) get compared with gradient boosting on the same features? Which evidence drives a match, as measured by the estimated logistic regression coefficients? And can a count model for the number of matches per entity (Poisson, negative binomial, multinomial logit) support the decision of how many candidates to accept?

The contributions are: (i) a reproducible pipeline that turns the raw records into 43 pair features after TF-IDF blocking; (ii) a comparison of four classifiers with grouped cross-validation, grid search and random search, scored on a lockbox split that was never touched while building the challenge system; (iii) an analysis of decision rules that convert pair probabilities into match sets, including an expected-F0.5 rule; and (iv) a count-data view of the problem, which shows that match counts in this data are under-dispersed. All code, the executed notebook, the result tables and a dashboard are in our GitHub repository[1].

## II. Related Work

The probabilistic basis of record linkage goes back to Fellegi and Sunter [1], who scored record pairs by likelihood ratios of field agreement and accepted pairs above a threshold. Surveys by Elmagarmid et al. [3] and Christen [2] describe the now standard steps: normalisation, blocking, pair comparison, classification and evaluation. Blocking limits the quadratic number of comparisons; Papadakis et al. [4] review the main families, of which token and q-gram based similarity search is closest to our TF-IDF approach [5].

[1] https://github.com/Manthan-Vats/carcosa-entity-resolution. The challenge data belongs to the organisers; the repository holds only a 1,500-entity demo sample for the dashboard.

Pair comparison relies on string similarity. Cohen et al. [6] compared edit-based and token-based metrics for names and found hybrid token measures strongest, while the Jaro-Winkler metric [7] remains common for short names. Bilenko and Mooney [8] showed that learning how to combine similarities with a classifier beats hand-tuned rules. Magellan [9] packaged this features-plus-classifier recipe, where random forests and logistic regression are typical choices. Deep models followed: DeepMatcher [10] and Ditto [11] embed both records with neural or pre-trained language models and gain most on textual, dirty data. They need GPUs and their decisions are harder to explain, so we keep to feature-based models, where gradient boosted trees [12], [13] are the strongest standard learner and logistic regression [14] offers direct inference on each piece of evidence.

The challenge metric is an F-measure averaged over entities, so the decision step matters as much as the probabilities. Lipton et al. [15] showed that the F1-optimal threshold of a calibrated classifier is half the optimal F1 score, and Jansche [16] trained models directly for expected F-measure. We use the expected-F idea per entity rather than one global threshold.

## III. Data and Preparation

### A. Records and labels

Each record has an identifier, a business name, an address and a country label. Source 1 is deduplicated; every Source 2/3 record matches at most one Source 1 entity, and a Source 1 entity can match zero, one or many records. The noise is typical of commercial data: legal suffix variants (Pvt/Private, Corp/Corporation), word order changes, typos, DBA names, landmark addresses ("near SBI ATM"), missing PIN codes and, for India, names written in Devanagari. Table I summarises missing addresses and non-Latin names in our sample.

### B. Normalisation

Names and addresses are lower-cased, "&" becomes "and", and text in Indic scripts is mapped to Latin with a token dictionary learned from matched training pairs (1,339 name tokens), backed by a generic transliteration library (`anyascii`) for unseen tokens. House and plot numbers are extracted into a separate field, and their leading zeros are stripped ("00604" $\rightarrow$ "604"). Missing values are not imputed: an empty address stays empty and is exposed to the models as a binary feature, and number features take the value $-1$ when either record has no number, together with an explicit missing-number indicator. No outliers were removed. Unusually long names (the 99th percentile is about 50 characters) are real records, and the length features capture them.

### C. Study sample and splits

Running the full pipeline on all records takes several hours, so the comparative study uses a 5% sample: 5% of the Source 1 entities, all their true matches, and 5% of the Source 2/3 records that match nobody. Because each Source 2/3 record belongs to at most one entity, the sample keeps the ratio of matched to unmatched records (4.7 pool records per entity in both). The sample has 110,341 entities and 625,153 records in total.

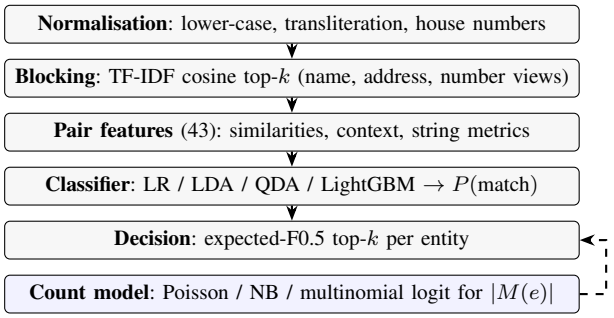



Fig. 1. Pipeline of the study. The count model is an alternative way to choose how many candidates to accept.

The split unit is the Source 1 entity, so all candidate pairs of an entity fall in the same split. We reuse the fold assignment of our challenge system: five development folds f0–f4 (16% of entities each), stratified by country and number of matches, and a 20% lockbox. Folds f1–f3 train the final models, f4 serves for early stopping and threshold choice, f0–f4 together form the five groups of the cross-validation used for tuning, and the lockbox is the test set.

## IV. Methodology

### A. Blocking

Comparing every Source 1 record with every pool record of the same country would mean about $10^{12}$ pairs on the full data. Blocking keeps a short candidate list per entity. We build L2-normalised TF-IDF vectors [5] for four views: name character 3-grams (robust to typos), name words, address words and house numbers. Very common terms are removed through a maximum document frequency (0.5% for character grams, 3% for name words). For a weighted combination of views $v$ with weights $w_v$, the joint score of a query $q$ and a pool record $r$ is

$$s(q, r) = \sum_{v} w_v \cos\big(\mathbf{x}_q^{(v)}, \mathbf{x}_r^{(v)}\big). \tag{1}$$

Four searches with different weights (address-heavy, name-heavy, number-aware, and name-only among records without an address) return their top 10–30 records, and the union forms the candidate set. Sparse top-$k$ matrix products make this fast enough for a laptop CPU.

### B. Pair features

Every candidate pair gets 43 features in four groups. *Blocking similarities*: the four view cosines, the joint score and the rank of the candidate in each search. *Context*: the gap between a candidate's score and the best score of the same entity, the gap to the best entity for the same candidate, ranks, and the number of candidates per entity and per pool record. These features let a pointwise classifier see the competition among candidates. *String similarity*: Levenshtein ratio, token-sort and token-set ratios, partial ratio and Jaro-Winkler on names [6],

[7]; token-set, token-sort and partial ratio on addresses; Jaccard overlap and first-number agreement of house numbers. *Record properties*: name and address lengths (log-transformed), empty address, Devanagari script and candidate source.

For the linear and discriminant models we drop one feature of each pair with absolute Pearson correlation above 0.95 on the training folds, which removes redundant copies such as the joint score and its rank. Gradient boosting uses all features.

### C. Classifiers

*Logistic regression* models the match probability of pair $i$ with feature vector $\mathbf{x}_i$ as

$$P(y_i = 1 \mid \mathbf{x}_i) = \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (2)$$

and the coefficients maximise the penalised log-likelihood

$$\ell(\boldsymbol{\beta}) = \sum_i \left[ y_i \log p_i + (1 - y_i) \log(1 - p_i) \right] - \frac{1}{2C} \|\boldsymbol{\beta}\|_2^2. \quad (3)$$

For inference we refit an unpenalised model by maximum likelihood (Newton-Raphson) on standardised inputs, so that $e^{\beta_j}$ is the odds ratio for a one standard deviation increase in feature $j$, and Wald $z$-tests give standard errors and $p$-values [14].

*Linear discriminant analysis* assumes Gaussian class densities with a shared covariance $\Sigma$ and classifies by

$$\delta_k(\mathbf{x}) = \mathbf{x}^\top \Sigma^{-1} \boldsymbol{\mu}_k - \tfrac{1}{2} \boldsymbol{\mu}_k^\top \Sigma^{-1} \boldsymbol{\mu}_k + \log \pi_k, \quad (4)$$

which gives a linear boundary [17], [18]. We tune Ledoit-Wolf style shrinkage of $\Sigma$ [19]. *Quadratic discriminant analysis* allows a covariance $\Sigma_k$ per class, which adds the term $-\frac{1}{2} \log |\Sigma_k|$ and makes the boundary quadratic; its regularisation parameter shrinks each $\Sigma_k$ towards the identity. *LightGBM* [13] fits an additive model of regression trees by gradient boosting on the log-loss [12] and serves as the non-linear reference, since it is the model of our challenge system.

### D. From probabilities to match sets

The challenge score of entity $e$ with predicted set $\hat{M}$ and true set $M$ is

$$F_{0.5}(e) = \frac{(1 + \beta^2)\, PR}{\beta^2 P + R}, \quad \beta = 0.5, \quad (5)$$

with precision $P = |\hat{M} \cap M| / |\hat{M}|$ and recall $R = |\hat{M} \cap M| / |M|$. An entity without matches scores 1 only when nothing is predicted, and the final score is the mean over entities. $\beta = 0.5$ weighs precision twice as much as recall. We compare three rules: a fixed threshold of 0.5; the threshold that maximises pair-level F1 on f4; and an *expected-F0.5 top-$k$* rule. With candidate probabilities sorted as $p_1 \geq p_2 \geq \ldots$, the rule picks

$$k^* = \arg\max_{k \geq 1} \frac{1.25 \sum_{j \leq k} p_j}{k + 0.25 \sum_j p_j}, \quad (6)$$

a ratio-of-expectations approximation of the expected F0.5, and predicts the empty set instead when $\prod_j (1 - p_j)$, the probability that the entity has no match among its candidates, is larger.

TABLE I
DATA QUALITY OF THE STUDY SAMPLE (NAME LENGTH: MEDIAN CHARACTERS)

| Src. | Country | Empty addr. % | Devanagari % | Name len. |
|---|---|---|---|---|
| S1 | India | 0.0 | 0.0 | 27 |
| S1 | US | 0.0 | 0.0 | 22 |
| S2 | India | 2.9 | 23.5 | 27 |
| S2 | US | 3.6 | 0.0 | 23 |
| S3 | India | 3.0 | 13.2 | 27 |
| S3 | US | 3.5 | 0.0 | 23 |

### E. Count models

Let $N_e = |M(e)|$ be the number of true matches of entity $e$. From eight entity-level summaries of the candidate list (counts of strong name and address agreements, the best joint and name scores, record lengths and country) we fit a Poisson GLM with log link, $\log \mathbb{E}[N_e] = \mathbf{z}_e^\top \boldsymbol{\gamma}$, and a negative binomial (NB2) model with $\mathrm{Var}(N_e) = \mu_e + \alpha \mu_e^2$, which relaxes the Poisson assumption that the variance equals the mean [20]. A multinomial logit over the classes $\{0, 1, 2, 3+\}$ treats the count as a category. The predicted mean also gives a decision rule: accept the $\mathrm{round}(\hat{\mu}_e)$ highest-ranked candidates.

### F. Validation and tuning

Hyperparameters are tuned by five-fold grouped cross-validation [21] on 15,000 development entities (about 790k pairs), with the predefined folds as groups so that no entity is split. Grid search covers the logistic regression penalty $C \in \{10^{-3}, \ldots, 10\}$ with and without class weighting, LDA shrinkage, and the QDA regularisation parameter. For LightGBM, random search [22] draws 12 settings of the number of leaves, learning rate, minimum leaf size, row and column subsampling and L2 penalty. The tuning score is average precision, because only about 6.5% of the pairs are matches and accuracy or ROC-AUC alone would hide differences in precision [23]. Pair-level results are reported as AUC-ROC, PR-AUC, F1, precision, recall, log-loss and Brier score; count models as MAE and RMSE. All models use scikit-learn [24], statsmodels [25] and LightGBM.

## V. EXPERIMENTS AND RESULTS

### A. Setup

Blocking on the study sample returned 5.80 million candidate pairs, 52.5 per entity, and kept 99.4% of the true matches. Positives make up 6.5% of the pairs. The final models are trained on 2.78M pairs (52,967 entities in f1–f3), validated on 0.93M pairs (f4) and tested on 1.14M pairs from 21,685 lockbox entities. The correlation filter removed five of the 43 features for the linear and discriminant models (address token-sort and partial ratio, the candidate address length, first-number agreement, and one gap feature). Everything ran on a 16-thread laptop CPU; the tuning stage took 19 minutes and the largest single fit, LightGBM, about two minutes.

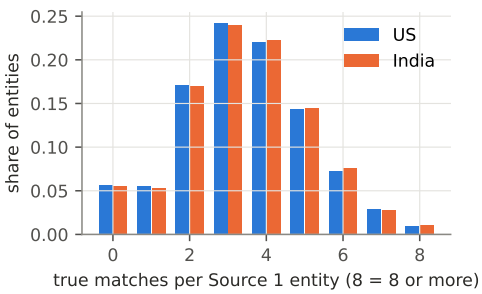



Fig. 2. Distribution of the number of true matches per Source 1 entity.

TABLE II
HYPERPARAMETER SEARCH WITH 5-FOLD GROUPED CROSS-VALIDATION

| Model | Search | Selected setting | CV AP (mean ± sd) |
|---|---|---|---|
| LR | grid | $C = 1$, weights: none | 0.9974 ± 0.0001 |
| LDA | grid | shrinkage: none | 0.9857 ± 0.0009 |
| QDA | grid | reg. param = 0.1 | 0.9917 ± 0.0004 |
| LightGBM | random | 85 leaves, lr 0.052 | 0.9991 ± 0.0001 |

### B. Exploratory analysis

The two countries look alike in structure. About 3–4% of Source 2/3 records have no address, while Source 1 addresses are always present (Table I). Devanagari appears only in Indian records: 23.5% of Source 2 names and 13.2% of Source 3 names. Fig. 2 shows the number of true matches per entity. The mode is three, 5.6% of entities have no match at all, and the distributions of the two countries almost coincide. The mean is 3.45 and the variance 2.88, a first sign that a Poisson model, whose variance equals its mean, will not fit well. Fig. 3 contrasts three similarity features. Matches pile up near the maximum, but a fair share of non-matches also have identical names (the right-hand spike of the name cosine), because chain stores and common trade names repeat across cities. The address similarity separates the classes better, which the coefficients below confirm.

### C. Tuning

Table II lists the selected settings. Logistic regression is flat across $C \geq 0.1$ (average precision 0.9973–0.9974) and class weighting did not help, so the default $C = 1$ without weights was kept. LDA worked best without shrinkage, which is expected with 790k rows and 38 features. QDA needed regularisation (0.1): without it the per-class covariance of the matches is close to singular. The random search for LightGBM chose 85 leaves and a learning rate of 0.052, and early stopping on f4 ended the final fit after 541 trees. Cross-validation standard deviations are at most 0.001, so the ranking of the models is stable across folds.

### D. Pair-level classification

All four models separate the classes well by AUC-ROC (Table III), with values between 0.998 and 1.000. That number hides the differences that matter, because with 1.07 million negatives even a false positive rate of 0.2% means two thousand wrong links. The precision-recall view and Fig. 4 are more telling. LightGBM reaches an F1 of 0.989 and makes 797 false positives, against 2,011 for logistic regression, 2,303 for LDA and 2,412 for QDA. Logistic regression is the best of the classical models on every metric (F1 0.979), and its log-loss of 0.008 shows well calibrated probabilities. LDA has the lowest recall (0.925), since its single shared covariance cannot fit the very different spread of the two classes. QDA ranks pairs well but its probabilities are extreme: its F1-optimal threshold is 0.987 and its log-loss five times that of logistic regression.

### E. What drives a match

The maximum likelihood fit on 250k training pairs reaches a McFadden pseudo-$R^2$ of 0.948. Fig. 5 plots the standardised coefficients. Three findings stand out. First, the strongest predictor is a context feature, `joint_gap_cand` ($\hat{\beta} = 5.73$, $p < 10^{-100}$): a record that resembles another Source 1 entity better than the current one is unlikely to belong to the current one. Since each Source 2/3 record has a single owner, competition between entities carries more information than any single similarity. Second, address agreement outweighs name agreement. The address token-set ratio has an odds ratio of 45 per standard deviation and house-number overlap one of 25, while name TF-IDF cosine has 1.45 once the string metrics are present. Address cosine is only borderline ($p = 0.015$) and the gap to the entity's best candidate is not significant ($p = 0.40$). Third, a Devanagari candidate is less likely to match at the same similarity level (odds ratio 0.67), which suggests that transliteration makes non-matching names look more alike than they are. The large positive coefficients of `num_missing` and `c_addr_empty` should be read together with the features they flag: they offset the $-1$ code and the zero similarity that a missing value produces, and are not evidence that missing data indicates a match.

### F. Entity-level results

Table IV reports the challenge metric. LightGBM with the expected-F0.5 rule scores 0.9855, within 0.013 of the 0.9979 ceiling set by blocking. Logistic regression reaches 0.9716, QDA 0.9581 and LDA 0.9477. The decision rule matters in a model-dependent way. For the two calibrated models the expected-F0.5 rule is best: it adapts the number of accepted candidates to each entity and predicts nothing when no candidate is convincing. For logistic regression the F1 threshold (0.39) is even worse than 0.5, because it trades precision for recall while F0.5 rewards the opposite. For QDA the expected-F0.5 rule loses to the threshold, as the rule takes probabilities at face value and QDA's are badly calibrated. Indian entities score 0.1–2.3 points lower than US ones for every model, which matches the extra script and spelling noise seen in Table I.

### G. Overfitting and underfitting

Fig. 6 shows both failure modes. On the left, a deliberately flexible LightGBM (255 leaves, five samples per leaf) drives

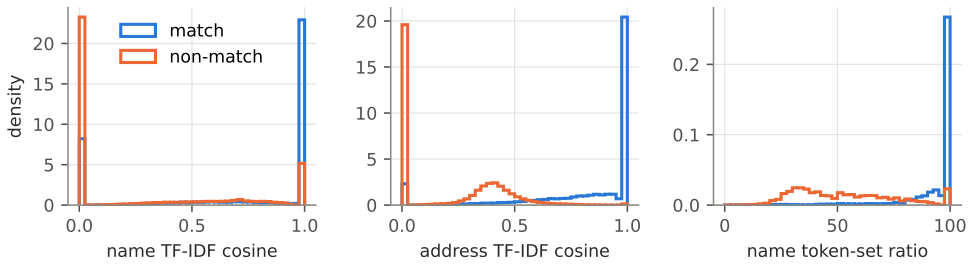



Fig. 3. Densities of three pair features for matching and non-matching candidates (400k random pairs).

TABLE III
PAIR-LEVEL RESULTS ON THE LOCKBOX (1.14M PAIRS, F1 THRESHOLD CHOSEN ON F4)

| Model | AUC-ROC | PR-AUC | F1 | Prec. | Rec. | Log-loss | Brier |
|---|---|---|---|---|---|---|---|
| LR | 0.9998 | 0.9972 | 0.9786 | 0.9733 | 0.9840 | 0.0081 | 0.0022 |
| LDA | 0.9983 | 0.9845 | 0.9460 | 0.9676 | 0.9253 | 0.0760 | 0.0066 |
| QDA | 0.9987 | 0.9884 | 0.9690 | 0.9677 | 0.9704 | 0.0386 | 0.0042 |
| LightGBM | **1.0000** | **0.9993** | **0.9885** | **0.9893** | **0.9878** | **0.0041** | **0.0011** |

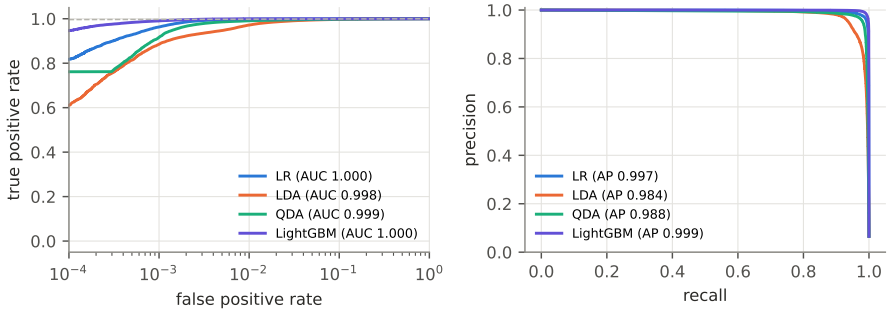



Fig. 4. ROC curves (log-scaled false positive rate) and precision-recall curves on the lockbox.

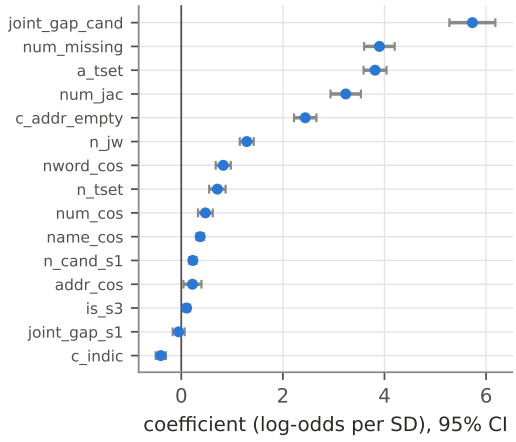



Fig. 5. Logistic regression coefficients on standardised features with 95% Wald intervals.

TABLE IV
ENTITY-LEVEL MACRO F0.5 ON THE LOCKBOX (21,685 ENTITIES); US/INDIA USE THE EXPECTED-F0.5 RULE

| Model | $p \geq 0.5$ | F1 thr. | Exp.-F0.5 | US | India |
|---|---|---|---|---|---|
| LR | 0.9703 | 0.9683 | 0.9716 | 0.9720 | 0.9710 |
| LDA | 0.9473 | 0.9475 | 0.9477 | 0.9564 | 0.9348 |
| QDA | 0.9476 | 0.9581 | 0.9512 | 0.9604 | 0.9374 |
| LightGBM | 0.9840 | 0.9843 | **0.9855** | 0.9869 | 0.9834 |
| Perfect classifier on the blocked candidates: 0.9979 | | | | | |

the training log-loss to $4 \times 10^{-8}$, while the validation log-loss reaches its minimum of 0.0054 at round 87 and then nearly doubles to 0.0102 by round 800. Early stopping and the tuned leaf size of 97 avoid this. On the right, a single decision tree underfits at small depths (average precision 0.81 at depth one on both sets), peaks on validation at depth eight (0.988), and overfits beyond it: at depth 24 and above the tree reproduces the training data exactly while validation precision drops to 0.951.

### H. Count models

The Poisson GLM finds sensible effects: the count rises with the number of candidates that agree strongly on both name and address ($\hat{\gamma} = 0.60$ on the log scale) and with

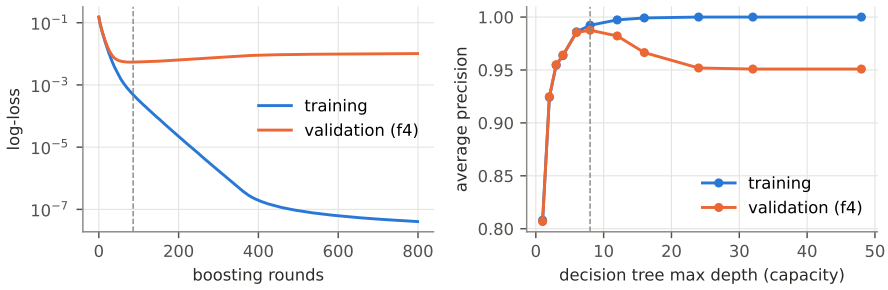



Fig. 6. Left: training and validation log-loss of an over-flexible LightGBM; the dashed line marks the validation minimum. Right: average precision of a decision tree against its maximum depth.

TABLE V
COUNT MODELS FOR THE NUMBER OF MATCHES PER ENTITY (LOCKBOX). THE MULTINOMIAL LOGIT PREDICTS CAPPED CLASSES, SO ITS MAE/RMSE ARE ON THE 0–3 SCALE

| Model | AIC | MAE | RMSE | Acc. (0/1/2/3+) |
|---|---|---|---|---|
| Poisson GLM | 288,006 | 0.713 | 0.949 | 0.763 |
| Negative binomial | 288,008 | 0.713 | 0.949 | 0.763 |
| Multinomial logit | 77,703 | 0.211 | 0.506 | 0.808 |

the best name similarity, and is about 10% lower for Indian entities at equal evidence. Its fit, however, shows strong under-dispersion: the Pearson statistic divided by the residual degrees of freedom is 0.31, far below the value of one implied by the model. The negative binomial can only add variance, so its dispersion parameter collapses to $\hat{\alpha} \approx 2 \times 10^{-8}$ and it reproduces the Poisson fit almost exactly (Table V). The source of under-dispersion is plain: most entities have two to five matches and few have none or many, so the counts are more regular than independent events would be. Treating the count as a category works better. The multinomial logit over $\{0, 1, 2, 3+\}$ classifies 80.8% of the lockbox entities correctly, against 76.3% for the rounded Poisson mean on the same classes.

Used as a decision rule, the count model falls well short. Accepting the top $\mathrm{round}(\hat{\mu})$ LightGBM candidates gives a macro F0.5 of only 0.860, against 0.9855 for the expected-F0.5 rule. A mean absolute count error of 0.71 is small for a regression, but every unit of error is a missed or wrong link. The pair probabilities already contain sharper information about each entity than eight summary statistics do.

### I. The full challenge system

Table VI traces our challenge entry on the full data, where each entity competes with 20 times more look-alikes. The largest gains came from blocking, not from the classifier: union blocking and transliteration raised the recall ceiling from 0.888 to 0.935, and a name-word TF-IDF view raised it to 0.973, lifting the dev score from 0.936 to 0.974 and the public leaderboard score from 0.902 to 0.959. Training the second stage on more data (r003) did not help. The two-stage design keeps the candidate file small (5.1 candidates per entity), which the challenge rewarded separately.

## VI. DISCUSSION

The comparison gives a clear order: gradient boosting first, then logistic regression, QDA and LDA. Logistic regression is the practical runner-up. It is 1.4 points behind LightGBM on the challenge metric, trains in eleven seconds instead of two minutes, and every coefficient can be read and tested. In a setting where links must be explained to a reviewer or an auditor, that trade may be worth making. The discriminant models suffer from the Gaussian assumption. Similarity scores are bounded and pile up at their maximum (Fig. 3), which is far from normal, so LDA and QDA lose recall and calibration respectively.

Two lessons carry beyond this data. First, a metric that averages over entities must be optimised over entities: a fixed pair threshold, even one tuned for F1, leaves score on the table, while the expected-F0.5 rule needs only calibrated probabilities. Second, checking the dispersion of a count before choosing between Poisson and negative binomial is not a formality. Here the usual remedy for a poor Poisson fit, the negative binomial, cannot help, because the problem is too little variance rather than too much.

### A. Ethical considerations

Business records are less sensitive than personal data, but ER systems in practice also link sole proprietors, whose business name is often a person's name and whose address is a home. A false merge can attach one trader's complaints or credit history to another, and the harm falls on the smaller party, who may not know the link exists. For that reason we favour a precision-weighted metric, keep the model's evidence inspectable through the logistic regression coefficients and the dashboard, and recommend human review for low-confidence links before they change any account. The model performs slightly worse on Indian records, which carry more script and spelling variation; a deployed system should monitor error rates per country and script so that one group of businesses does not bear most of the mistakes. The challenge data is used only for this academic study; apart from a small demo sample for the dashboard, it is not redistributed.

TABLE VI
DEVELOPMENT OF THE FULL CHALLENGE SYSTEM (ALL 1.76M DEV ENTITIES, SCORED ON FOLD F0)

| Run | Change | Dev F0.5 | Recall cap | Cands/S1 | Public LB |
|---|---|---|---|---|---|
| r001 | TF-IDF top-15, LightGBM prune + pair model | 0.9364 | 0.8881 | 4.27 | 0.902 |
| r002 | union blocking, transliteration, sibling features | 0.9546 | 0.9346 | 5.59 | 0.927 |
| r003 | r002 with stage B on more data | 0.9535 | 0.9346 | 5.59 | – |
| r004 | name-word view, cross-fitted stage A | 0.9742 | 0.9732 | 5.14 | 0.959 |

### B. Limitations

The study sample keeps the ratio of matched to unmatched records but has 20 times fewer records per country, so each entity faces fewer look-alike competitors. Blocking recall is therefore higher (99.4%) than on the full data (97.8%), and absolute scores are optimistic compared with the full challenge setting. The comparison between models is still fair, because all of them see the same candidates and features. The test countries of the challenge include France, which is absent from training; the lockbox only contains US and Indian records, so our results say nothing direct about transfer to a new country. Time series methods from the syllabus do not apply, since the records carry no timestamps.

## VII. CONCLUSION

We treated business entity resolution as a predictive modeling problem: TF-IDF blocking, 43 engineered pair features, four classifiers tuned with grouped cross-validation, an entity-level decision rule and count models for the number of matches. On a held-out lockbox of 21,685 entities, LightGBM reached a macro F0.5 of 0.9855 and logistic regression 0.9716, against a ceiling of 0.9979. The logistic coefficients show that address and house-number agreement and the competition between entities for a record matter more than name similarity. The match counts are under-dispersed, so the negative binomial model reduces to Poisson and a multinomial logit fits better, but no count model beats the expected-F0.5 rule as a way to choose matches. The full version of the pipeline reached 0.959 on the Amazon ML Challenge 2026 public leaderboard. Future work includes testing transfer to unseen countries such as France, ordinal or generalised Poisson models for under-dispersed counts, and calibrating QDA before using it in the expected-F rule.

## REFERENCES


[1] I. P. Fellegi and A. B. Sunter, "A theory for record linkage," *Journal of the American Statistical Association*, vol. 64, no. 328, pp. 1183–1210, 1969.

[2] P. Christen, *Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection*. Berlin, Germany: Springer, 2012.

[3] A. K. Elmagarmid, P. G. Ipeirotis, and V. S. Verykios, "Duplicate record detection: A survey," *IEEE Transactions on Knowledge and Data Engineering*, vol. 19, no. 1, pp. 1–16, 2007.

[4] G. Papadakis, D. Skoutas, E. Thanos, and T. Palpanas, "Blocking and filtering techniques for entity resolution: A survey," *ACM Computing Surveys*, vol. 53, no. 2, pp. 31:1–31:42, 2020.

[5] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.

[6] W. W. Cohen, P. Ravikumar, and S. E. Fienberg, "A comparison of string distance metrics for name-matching tasks," in *Proc. IJCAI Workshop on Information Integration on the Web*, 2003, pp. 73–78.

[7] W. E. Winkler, "String comparator metrics and enhanced decision rules in the Fellegi-Sunter model of record linkage," in *Proc. Section on Survey Research Methods, American Statistical Association*, 1990, pp. 354–359.

[8] M. Bilenko and R. J. Mooney, "Adaptive duplicate detection using learnable string similarity measures," in *Proc. 9th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2003, pp. 39–48.

[9] P. Konda, S. Das, P. Suganthan G. C., A. Doan *et al.*, "Magellan: Toward building entity matching management systems," *Proceedings of the VLDB Endowment*, vol. 9, no. 12, pp. 1197–1208, 2016.

[10] S. Mudgal, H. Li, T. Rekatsinas, A. Doan, Y. Park, G. Krishnan, R. Deep, E. Arcaute, and V. Raghavendra, "Deep learning for entity matching: A design space exploration," in *Proc. ACM SIGMOD Int. Conf. Management of Data*, 2018, pp. 19–34.

[11] Y. Li, J. Li, Y. Suhara, A. Doan, and W.-C. Tan, "Deep entity matching with pre-trained language models," *Proceedings of the VLDB Endowment*, vol. 14, no. 1, pp. 50–60, 2020.

[12] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.

[13] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems 30*, 2017, pp. 3146–3154.

[14] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed. Hoboken, NJ, USA: Wiley, 2013.

[15] Z. C. Lipton, C. Elkan, and B. Naryanaswamy, "Optimal thresholding of classifiers to maximize F1 measure," in *Machine Learning and Knowledge Discovery in Databases (ECML PKDD), LNCS 8725*, 2014, pp. 225–239.

[16] M. Jansche, "Maximum expected F-measure training of logistic regression models," in *Proc. Human Language Technology Conf. and Conf. Empirical Methods in Natural Language Processing*, 2005, pp. 692–699.

[17] R. A. Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, vol. 7, no. 2, pp. 179–188, 1936.

[18] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.

[19] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.

[20] A. C. Cameron and P. K. Trivedi, *Regression Analysis of Count Data*, 2nd ed. Cambridge, U.K.: Cambridge University Press, 2013.

[21] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in *Proc. 14th Int. Joint Conf. Artificial Intelligence (IJCAI)*, 1995, pp. 1137–1143.

[22] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, pp. 281–305, 2012.

[23] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, p. e0118432, 2015.

[24] F. Pedregosa, G. Varoquaux, A. Gramfort *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[25] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with Python," in *Proc. 9th Python in Science Conf.*, 2010, pp. 92–96.